Ch 6.3: Dimension Reduction - PCA

Lecture 19 - CMSE 381

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Mon, Oct 13, 2025

Announcements

Last time:

- Shrinkage: Ridge and Lasso

This lecture:

- PCA

Announcements:

- Exam #2 on Wednesday!
  - Bring 8.5x11 sheet of paper
  - Handwritten both sides
  - Anything you want on it, but must be your work
  - You will turn it in
  - Non-internet calculator
- Project: by Exam # 2
  - project partner
  - ideas about what method to use

CMSE381_F2025_Schedule : Schedule

	M	9/22	***Project Day & Review***		
	W	9/24	**Midterm #1**		
12	F	9/26	Leave one out CV	5.1.1, 5.1.2	
13	M	9/29	k-fold CV	5.1.3	
14	W	10/1	More k-fold CV	5.1.4-5	
15	F	10/3	k-fold CV for classification	5.1.5	
16	M	10/6	Subset selection	6.1	
17	W	10/8	Shrinkage: Ridge	6.2.1	
18	F	10/10	Shrinkage: Lasso	6.2.2	HW #4 Due Sun 10/12
19	M	10/13	PCA	6.3	
20	W	10/15	PCR	6.3	
	F	10/17	***Review***		
	M	10/20	Fall Break		
	W	10/22	**Midterm #2**		
21	F	10/24	Polynomial & Step Functions	7.1-7.2	HW #5 Due Sun 10/28
22	M	10/27	Step Functions; Basis functions; Start Splines	7.2-7.4	
23	W	10/29	Regression Splines	7.4	

Section 1

Last time

Goal

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

- Ridge regression
- Lasso

Shrinkage

Find β to minimize:

Least Squares:

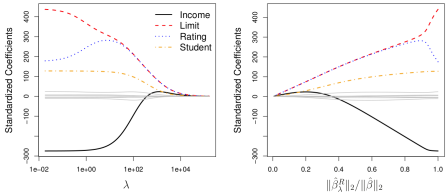
$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

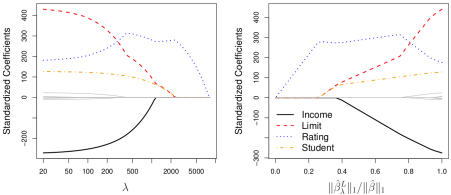
Ridge:

$$RSS + \sum_{j=1}^{p} \beta_j^2$$

The Lasso:

$$RSS + \sum_{j=1}^{p} |\beta_j|$$





Also, find best choice of λ or s using CV

Section 2

Dimension Reduction

Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

- The goal is for $M \ll p$
- Need to figure out good φ to do this

An example or two

Two dimensions down to 1.....

- $$Z_1 = X_1 + 3 \cdot X_2$$
- Matrix version:

$$(Z_1) = \begin{pmatrix} X_1 & X_2 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

- Example data:

X_1	X_2	Z_1
0	1	4
3	4	17
1	1	4
-1	0	-1

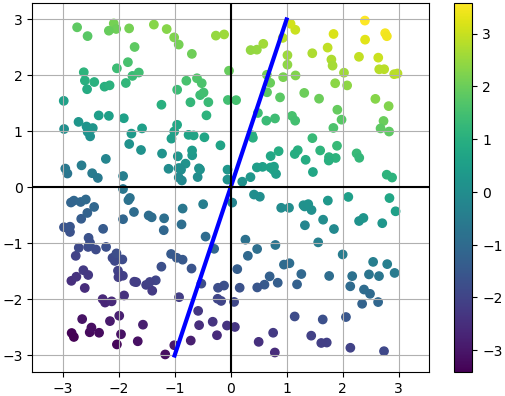
Create an example with just three dimensions down to two

- $\varphi = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \\ 0 & 1/2 \end{pmatrix}$

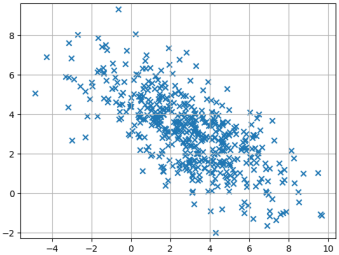
$$Z_1 = X_1 + 0 \cdot X_2 + 0 \cdot X_3$$
$$Z_2 = 0 \cdot X_1 + \tfrac{1}{2} X_2 + \tfrac{1}{2} X_3$$

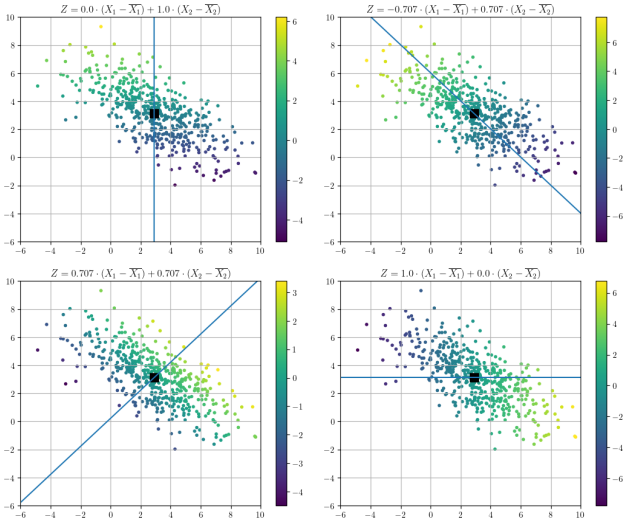
Geometric interpretation

- Get the φ to unit norm
- Example: $Z_1 = \frac{1}{\sqrt{10}} X_1 + \frac{3}{\sqrt{10}} X_2$
- Point out that the Z value for the point is how far along the point projected down onto the line
- projection on a line

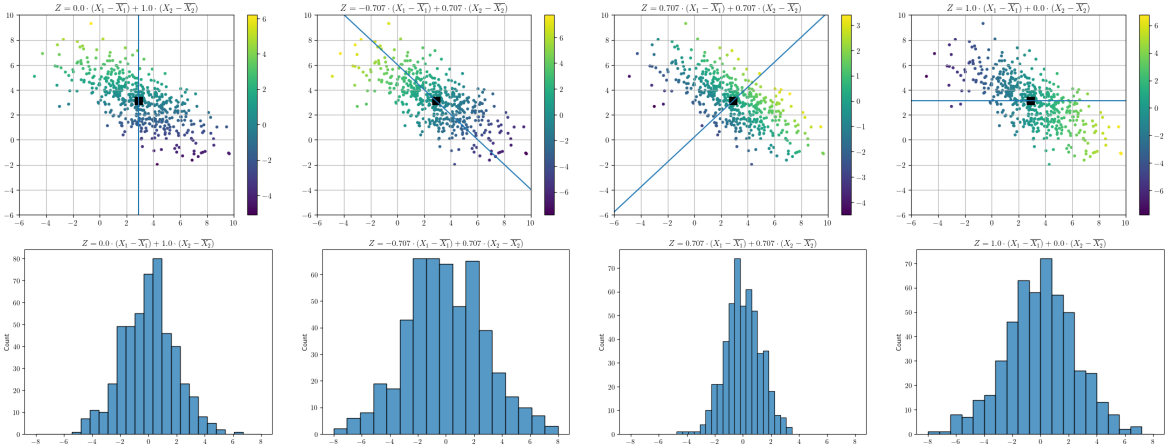


Different projections





Histograms of Z values



The goal

- Find good φ's for some $M \ll p$
- Fit regression model on Z_i's using least squares

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

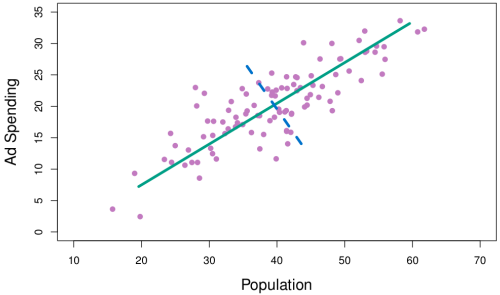
- in this notation, θs replace βs

- Dimension reduction comes from the fact that we're fitting models on a smaller number of variables
- Next two subsection: ways to find φs: PCA and partial least squares
- Today.... just PCA?

Section 3

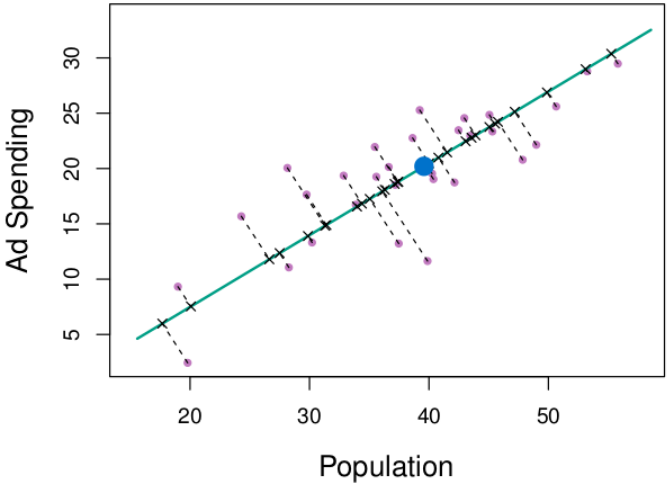
PCA

An example dataset



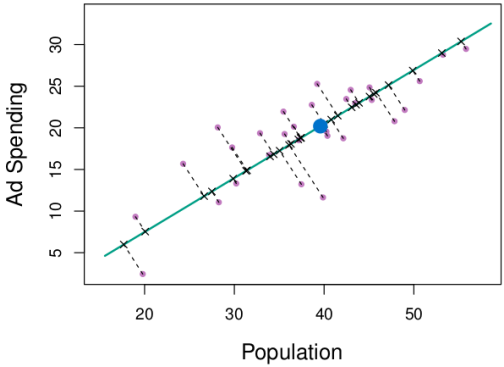
- Population size in tens of thousands of people
- Ad spending for a company in thousands of dollars
- Note: These are both input variables
- 100 data points
- By eye: green line is the direction of most variability, called the principal component
- Meaning if we project observations onto the line, then the projected observations would have the largest possible variance
-

Projection onto first PC



$$Z_1 = 0.839 \cdot (\mathtt{pop} - \overline{\mathtt{pop}}) + 0.544 \cdot (\mathtt{ad} - \overline{\mathtt{ad}})$$

What does it mean to have the highest variance



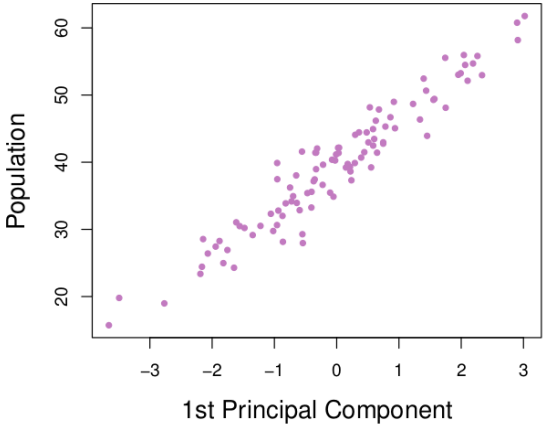
- Of every linear combo of pop and ad, this one has the highest variance
- Maximizes $\mathrm{Var}(\varphi_{1,1}(pop - \overline{pop}) + \varphi_{2,1}(ad - \overline{ad}))$
- Require $\varphi_{11}^2 + \varphi_{21}^2 = 1$, otherwise, just make φ super big to blow up variance

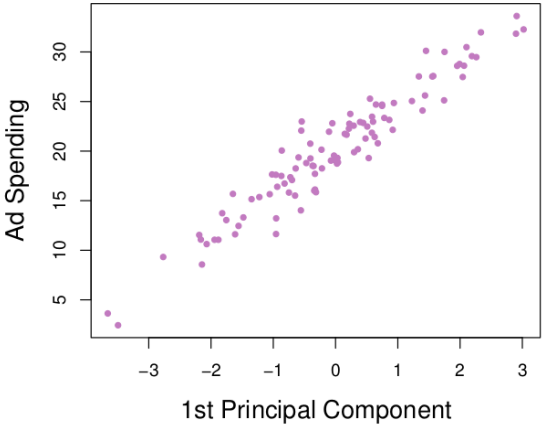
Toy for learning PCA

https://www.desmos.com/calculator/gvmq07pg1k

Don't forget to click step 2a to show points for PC2.

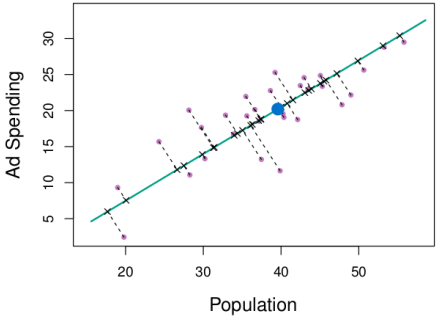
Principal component scores

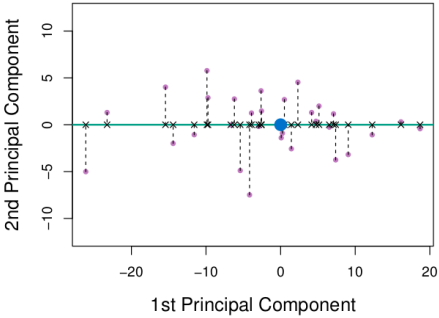




$$z_{i1} = 0.839 \cdot (\mathrm{pop}_i - \overline{\mathrm{pop}}) + 0.544 \cdot (\mathrm{ad}_i - \overline{\mathrm{ad}})$$

Another view





- Think of this as rotated for now, 2nd PC on next slide

- The score can be thought of as relation from the mean
- $z_{i1} < 0$ has below avg pop size as below avg ad spending

The other principal components

- Z_2 is linear combo of variables uncorrelated to Z_1
- Find the one that explains the most variance
- Result: requires Z_2 to be orthogonal or perpendicular

- How does your compy do it? Fun with matrices, outside the scope of class, but you should be able to interpret the output

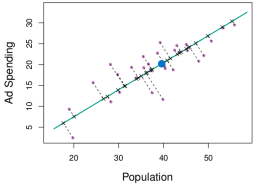
Draw the idea in 3D with circle of directions around a line in 3D

Do PCA with Penguins

TL;DR

PCA

- Unsupervised dimensionality reduction
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



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